Transitions Systems

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Transition Graphs

The semantics of a Boolean network f of dimension n are given by the transition system $\left(S = \mathbb{B}^n, \underset{f}{\longrightarrow}\right)$.

The transition system gives rise to a directed graph (V, E) where $V = S = \mathbb{B}^n$ and $E \subseteq V \times V = \underset{f}{\longrightarrow}$.

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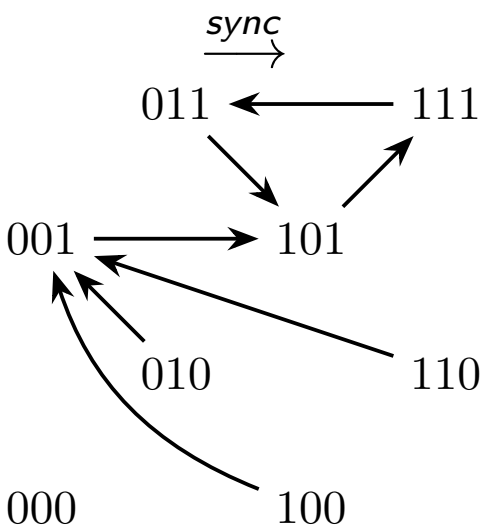
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$f_1(\mathbf{x}) = \mathbf{x}_3 \wedge (\neg\mathbf{x}_1 \vee \neg\mathbf{x}_2)$, $f_2(\mathbf{x}) = \mathbf{x}_1 \wedge \mathbf{x}_3$, $f_3(\mathbf{x}) = \mathbf{x}_1 \vee \mathbf{x}_2 \vee \mathbf{x}_3$

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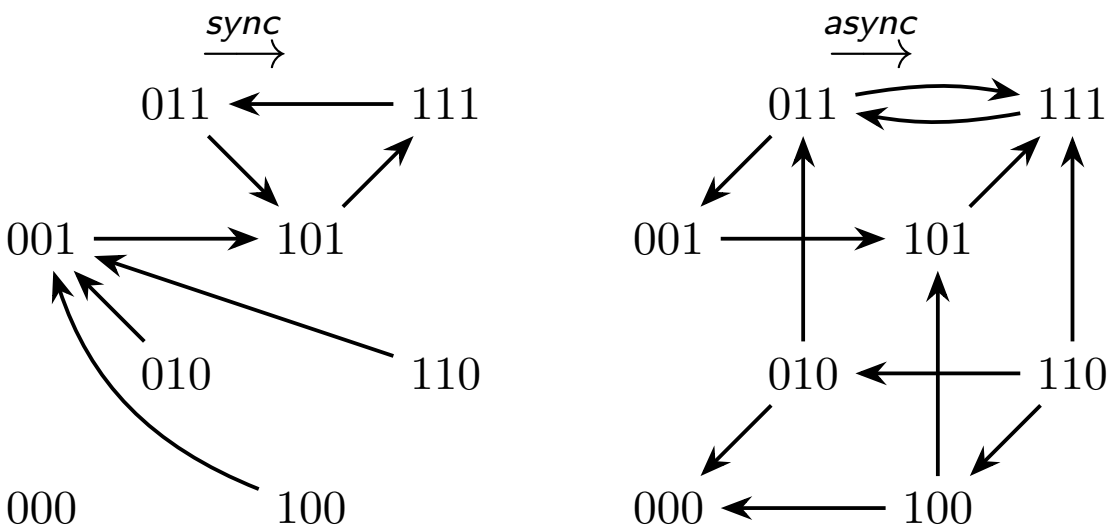


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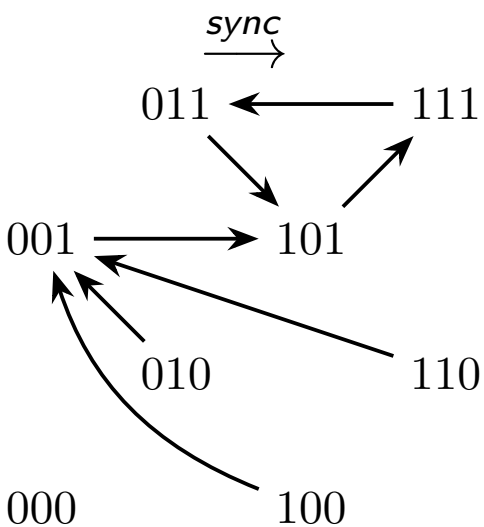


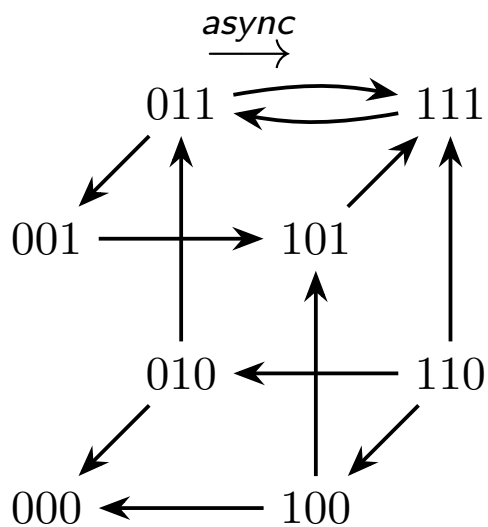
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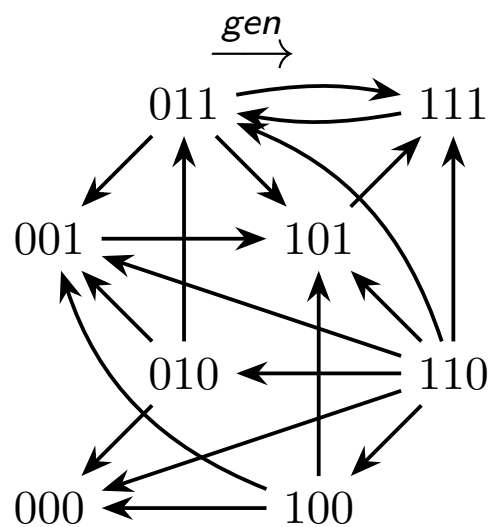
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Reachability

A **trace** $\sigma = (\mathbf{x}^0, \mathbf{x}^1, \mathbf{x}^2, \dots)$ is an (infinite) sequence of configurations such that for all $k > 0$, $\mathbf{x}^{k-1} \to \mathbf{x}^k$.

We often write $\sigma = \mathbf{x}^0 \to \mathbf{x}^1 \to \mathbf{x}^2 \to \dots$.

Each trace represents a possible behaviour of the model, starting from the **initial state** $\mathbf{x}^0$.

A configuration $\mathbf{y}$ is **reachable** from $\mathbf{x}$ if and only if there exists a trace $\sigma = (\mathbf{x} = \mathbf{x}^0, \mathbf{x}^1, \mathbf{x}^2, \dots)$ and an integer $k \geq 0$ such that $\mathbf{x}^k = \mathbf{y}$.

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Alternatively, let $\to^* \subseteq \mathbb{B}^n \times \mathbb{B}^n$ be the reflexive and transitive closure of the semantics relation $\to$.

Then $\mathbf{y}$ is reachable from $\mathbf{x}$ if and only if $\mathbf{x} \to^* \mathbf{y}$.

Trap Sets

A trap set is a non-empty set of configurations $\emptyset \neq T \subseteq \mathbb{B}^n$ which satisfies the following property:

$$\forall \mathbf{x} \in T, \forall \mathbf{y} \in \mathbb{B}^n, \mathbf{x} \rightarrow^* \mathbf{y} \Rightarrow \mathbf{y} \in T$$

Let $\mathcal{T}$ denote the set of all trap sets of a given BN.

Let $T, T' \in \mathcal{T}$ be two trap sets. We say T is smaller than T' if and only if it is included in T', $T \subseteq T'$.

For each configuration $\mathbf{x} \in \mathbb{B}^n$, let $[\mathbf{x}]_T \triangleq \{\mathbf{y} \in \mathbb{B}^n \mid \mathbf{x} \rightarrow^* \mathbf{y}\}$ denote the smallest trap set containing $\mathbf{x}$.

A trap set T is minimal, if $\forall T' \in \mathcal{T}$, $T \not\supset T'$.

$[y]_T \subseteq [x]_T$

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Trap set properties

Closed by non-empty intersection: $T, T' \in \mathcal{T}$ such that $T \cap T' \neq \emptyset$, then $T \cap T' \in \mathcal{T}$.
Minimal trap sets are "strongly connected": $T \in \mathcal{T}$ minimal, then $\forall \mathbf{x}, \mathbf{y} \in T, \mathbf{x} \rightarrow^* \mathbf{y}$.

$x \in T \cap T'$ $\quad [x]_{\mathcal{T}} \subseteq T$, $[x]_{\mathcal{T}} \subseteq T'$, $[x]_{\mathcal{T}} \subseteq T \cap T'$

$x, y \in T$ $\quad x \not\rightarrow^* y$ $\quad [x]_{\mathcal{T}} \subseteq T$, $y \notin [x]_{\mathcal{T}}$, $y \in T$, $[x]_{\mathcal{T}} \subsetneq T$

Attractors

bottom

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An attractor is a non-empty set of configurations $\emptyset \neq A \subseteq \mathbb{B}^n$ which satisfies the following properties:

1. $\forall \mathbf{x} \in A, \forall \mathbf{y} \in \mathbb{B}^n, \mathbf{x} \rightarrow^* \mathbf{y} \Rightarrow \mathbf{y} \in A$ (is a trap set);
2. $\forall \mathbf{x}, \mathbf{y} \in A, \mathbf{x} \rightarrow^* \mathbf{y}$ (is minimal);

$\forall x \in A, [x]_f \subseteq A$

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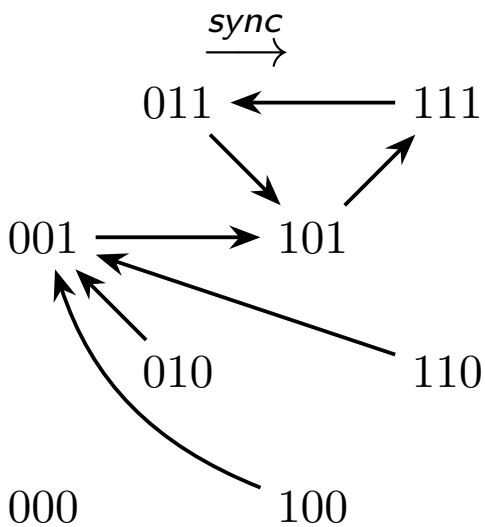
Types of attractors:

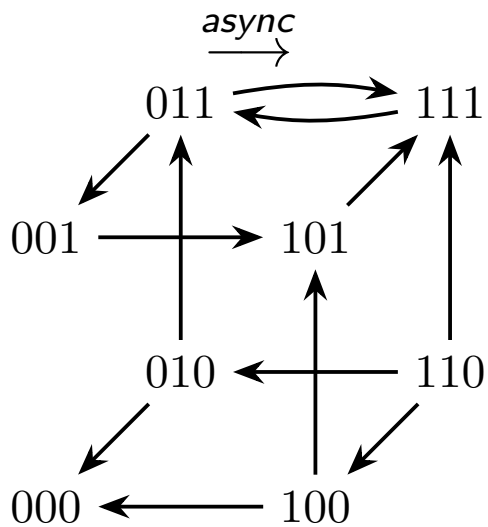
- **Fixed points** (singletons), $\forall \mathbf{x}, \mathbf{y} \in A, \mathbf{x} = \mathbf{y}$; $f(x) = x$
- **Cyclic attractors** (or limit cycles), consisting of at least two distinct configurations, $\exists \mathbf{x}, \mathbf{y} \in A, \mathbf{x} \neq \mathbf{y}$;

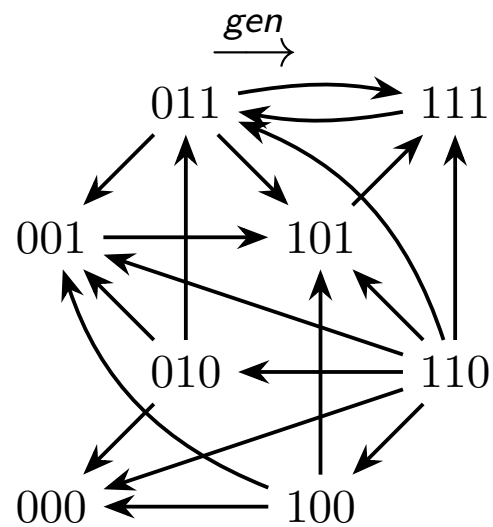
A configuration which is not part of any attractor is called **transient**.

Attractors – Example

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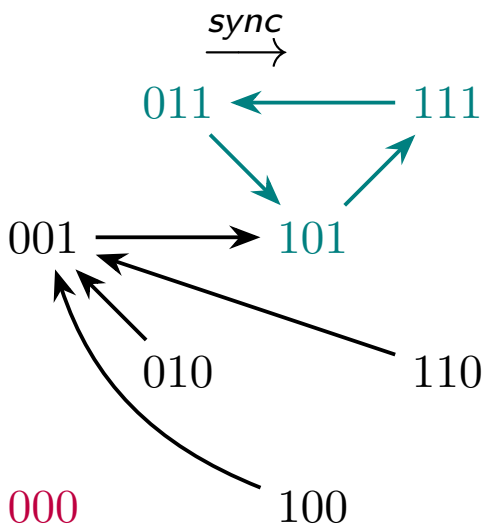


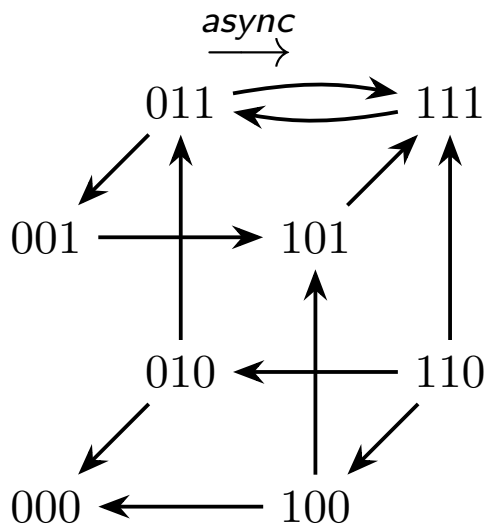


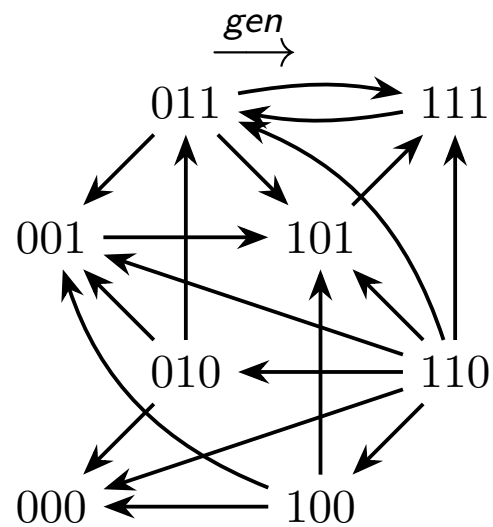


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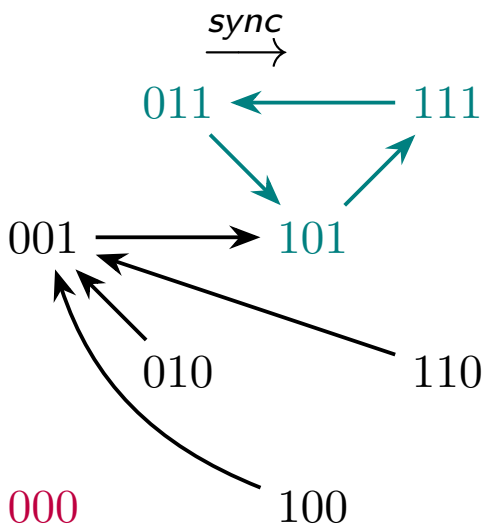


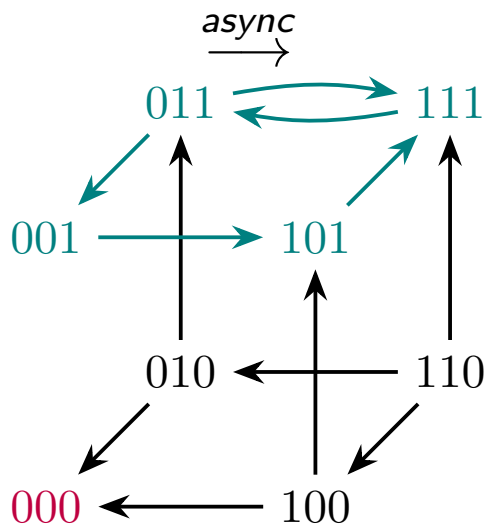
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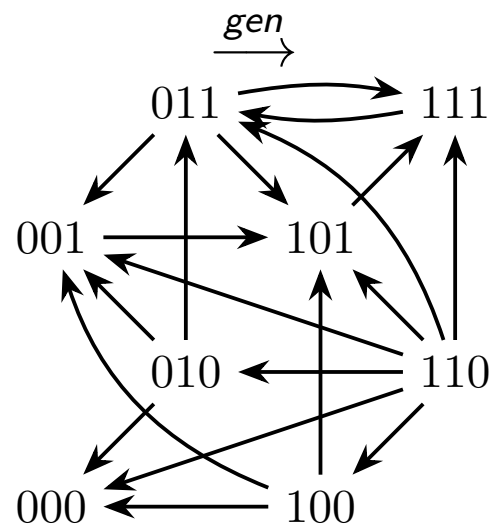
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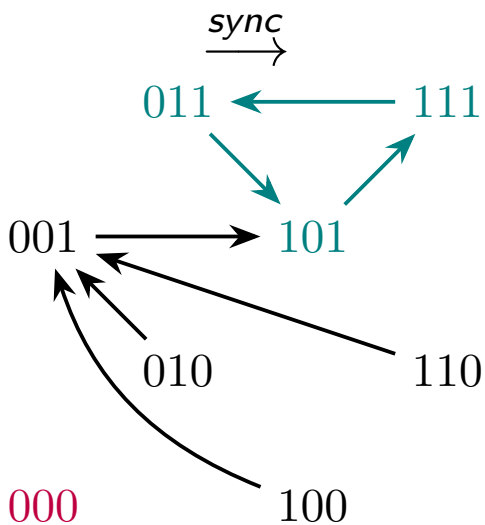


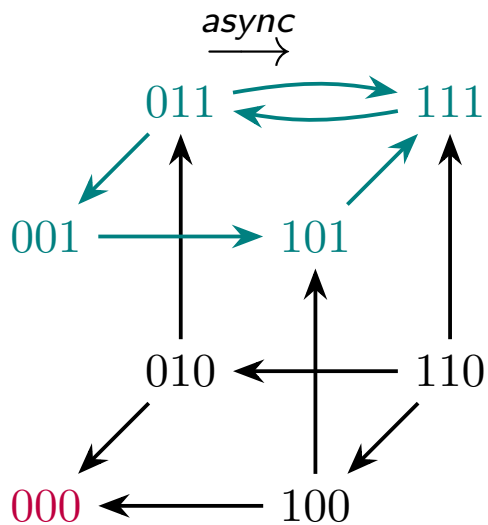
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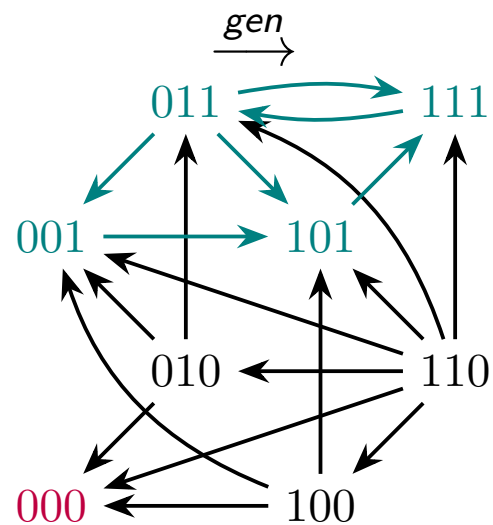
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Basins of Attraction

The **weak basin of attraction** of an attractor A is a set of configurations $\mathcal{WB}(A)$ defined as:

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where $\mathcal{A}$ is the set of all attractors of the BN.

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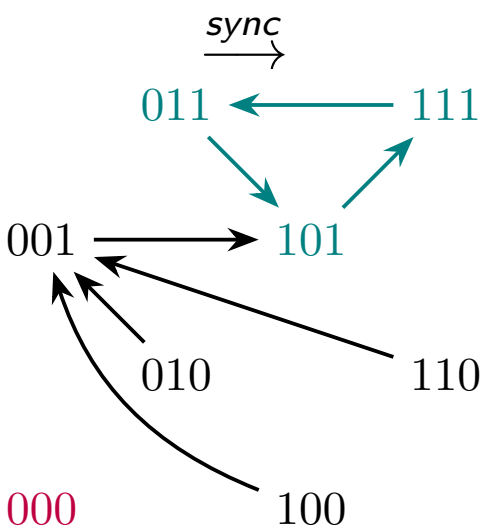
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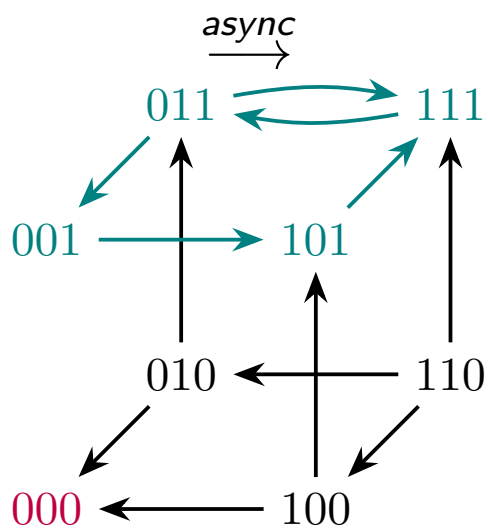
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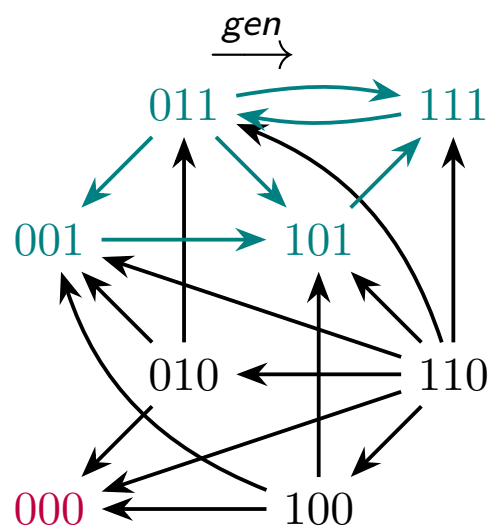
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Trap Spaces

A **subspace** of $\mathbb{B}^n$ is a partial function $d\colon \{1,\ldots,n\} \hookrightarrow \mathbb{B}$.
The variables on which d is defined, $D = \mathit{range}(d) \subseteq \{1,\ldots,n\}$, are called **fixed**.
The variables which are not fixed, $F = \{1,\ldots,n\} \setminus D$, are called **free**.
We denote the subspaces by vectors $\mathbf{d} \in \{0,1,*\}^n$ such that for each $i \in \{1,\ldots,n\}$:

$$\mathbf{d}_i \triangleq \begin{cases} 0 & \text{if } i \in D \text{ and d(i)=0} \\ 1 & \text{if } i \in D \text{ and d(i)=1} \\ * & \text{if } i \in F \end{cases}$$

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A **trap space** is a subspace which is also a trap set.

$\forall x \in d \;,\; f(x) \in d$